

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5303B

5 Credits: 4

Program Elective

Source-grounded

Non-Relational Database

□ Basics of database is studied already

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5303B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5303B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	5303B
Course title	Non-Relational Database
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

29 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Basics of database is studied already
- To develop basic concepts and appreciate the applications of database systems. Become familiar with the expanding role of database technology.
- systems. Understand the basic architecture and data models of a NoSQL
- To understand Emerging features in the area of database security
- To demonstrate the basic concepts of transaction processing and concurrency

Course outcomes

CO	Official outcome	Study level
CO1	Databases Demonstrate the features of indexing and 17 Understand	See official syllabus
CO2	hashing in database applications To Demonstrate the basic concepts of 15 Understand	See official syllabus
CO3	transaction processing and concurrency control Identify security and integrity threads in 14 Understanding	See official syllabus
CO4	database	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 3 3 2
CO2	CO2 3 2 3 3 3
CO3	CO3 3 2 3 3 2

Row	Extracted official mapping
CO4	CO4 3 3 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain and compare different types of NoSQL Databases	8	As specified
Module 2	Demonstrate the features of indexing and hashing in database applications	7	As specified
Module 3	concurrency control	7	As specified
Module 4	Identify security and integrity threads in database	7	As specified

MODULE 1

Explain and compare different types of NoSQL Databases

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain and compare different types of NoSQL Databases
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key

NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, nosql data model: aggregate models- 2 understanding document data model- key should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key definition/meaning . . . , steps, application Technical terms English-

▼ Key-Value Data Model Columnar Data Model

Key-Value Data Model Columnar Data Model is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Key-Value Data Model Columnar Data Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, key-value data model columnar data model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Key-Value Data Model Columnar Data Model means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Key-Value Data Model Columnar Data Model
definition/meaning .
steps, application . Technical terms English-
.

▼ Graph Based Data Model

Graph Based Data Model is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Graph Based Data Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, graph based data model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Graph Based Data Model means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഗ്രാഫ് ബേസ്ഡ് ഡാറ്റാ മോഡൽ (Graph Based Data Model) ന്റെ definition/meaning, steps, application ന്റെ വിവരങ്ങൾ നൽകുക. Technical terms English-ൽ നൽകുക.

▼ NoSQL system ways to handle big data 2 Understanding problems

NoSQL system ways to handle big data 2 Understanding problems is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify NoSQL system ways to handle big data 2 Understanding problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, nosql system ways to handle big data 2 understanding problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What NoSQL system ways to handle big data 2 Understanding problems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- NoSQL system ways to handle big data 2 Understanding problems definition/meaning steps, application Technical terms English-

▼ Moving Queries to data, not data to the query

Moving Queries to data, not data to the query is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Moving Queries to data, not data to the query using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, moving queries to data, not data to the query should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Moving Queries to data, not data to the query means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Moving Queries to data, not data to the query
definition/meaning.
steps, application
Technical terms English-

▼ Hash rings to distribute the data on clusters

Hash rings to distribute the data on clusters is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Hash rings to distribute the data on clusters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, hash rings to distribute the data on clusters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Hash rings to distribute the data on clusters means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Hash rings to distribute the data on clusters
 definition/meaning .
 , steps, application . Technical terms English- .

▼ Replication to scale reads

Replication to scale reads is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Replication to scale reads using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, replication to scale reads should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Replication to scale reads means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റൂപ്ലിക്കേഷൻ Replication to scale reads റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ definition/meaning റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ. റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ, steps, application റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ. Technical terms English-റൂപ്ലിക്കേഷൻ റൂപ്ലിക്കേഷൻ.

▼ **Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes.**

Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. is an official topic in Module 1 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, database distributed queries to data nodes. 2 understanding nosql data model: aggregate models- document data model- key-value data model columnar data model, graph based data model,, nosql system ways to handle big data problems, moving queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, database distributed queries to data nodes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Database distributed queries to Data nodes. 2 Understanding NoSQL Data model: Aggregate Models- Document Data Model- Key-Value Data Model Columnar Data Model, Graph Based Data Model,, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to Data nodes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

MODULE 2

Demonstrate the features of indexing and hashing in database applications

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title:
Demonstrate the features of indexing and hashing in database applications
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Data Storage and Indexes- File organizations 2 Understanding.

Data Storage and Indexes- File organizations 2 Understanding. is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Data Storage and Indexes- File organizations 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, data storage and indexes- file organizations 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Data Storage and Indexes- File organizations 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Data Storage and Indexes- File organizations 2 Understanding. ☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ Primary indexing

Primary indexing is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Primary indexing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, primary indexing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Primary indexing means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രാഥമിക ഇൻഡെക്സിംഗ് പ്രാഥമിക ഇൻഡെക്സിംഗ് definition/meaning പ്രാഥമിക ഇൻഡെക്സിംഗ്. പ്രാഥമിക ഇൻഡെക്സിംഗ് steps, application പ്രാഥമിക ഇൻഡെക്സിംഗ് പ്രാഥമിക ഇൻഡെക്സിംഗ്. Technical terms English-ൽ പ്രാഥമിക ഇൻഡെക്സിംഗ്.

▼ Secondary index structures

Secondary index structures is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Secondary index structures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, secondary index structures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Secondary index structures means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-☐☐ Secondary index structures

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Various index structures - hash-based

Various index structures - hash-based is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Various index structures - hash-based using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, various index structures - hash-based should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Various index structures - hash-based means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 Various index structures – hash-based
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ Dynamic hashing techniques

Dynamic hashing techniques is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Dynamic hashing techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, dynamic hashing techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Dynamic hashing techniques means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Dynamic hashing techniques

definition/meaning .
steps, application . Technical terms
English-
.

▼ Multi-level indexes,

Multi-level indexes, is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Multi-level indexes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, multi-level indexes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Multi-level indexes, means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇം Multi-level indexes, ഇം definition/meaning ഇം. ഇം steps, application ഇം. Technical terms English-ഇം ഇം.

▼ B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and

B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and is an official topic in Module 2 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures - hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, b and b+ trees(insertion and deletion) 2 applying data storage & indexing data storage and indexes: data storage and indexes - file organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, b and b+ trees(insertion and deletion) to demonstrate the basic concepts of transaction processing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- B and B+ trees(insertion and deletion) 2 Applying Data Storage & Indexing Data Storage and Indexes: Data Storage and Indexes - File organizations, primary, secondary index structures, various index structures – hash-based, dynamic hashing techniques, multi-level indexes, B and B+ trees(insertion and deletion) To Demonstrate the basic concepts of transaction processing and definition/meaning . , steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

concurrency control

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: concurrency control
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Transaction concept, transaction state

Transaction concept, transaction state is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Transaction concept, transaction state using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transaction concept, transaction state should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Transaction concept, transaction state means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-transaction concept, transaction state definition/meaning. Transaction steps, application. Technical terms English-transaction state.

▼ Implementation of atomicity and durability

Implementation of atomicity and durability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implementation of atomicity and durability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implementation of atomicity and durability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implementation of atomicity and durability means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Implementation of atomicity and durability
definition/meaning.
steps, application
Technical terms English-
.

▼ Concurrent executions

Concurrent executions is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Concurrent executions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, concurrent executions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Concurrent executions means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Concurrent executions
 definition/meaning . . . , steps, application Technical terms English-

▼ Serializability

Serializability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Serializability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, serializability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Serializability means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square\square$ Serializability $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ Recoverability

Recoverability is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recoverability using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recoverability should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recoverability means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി definition/meaning റീക്കവറബിലിറ്റി. റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി, steps, application റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി. Technical terms English-റീക്കവറബിലിറ്റി റീക്കവറബിലിറ്റി.

▼ Implementation of isolation

Implementation of isolation is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implementation of isolation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implementation of isolation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Implementation of isolation means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Implementation of isolation

പ്രധാനപ്പെട്ടതുകൾക്ക് definition/meaning നൽകുക. പദങ്ങൾക്ക് പ്രയോഗം, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

▼ Testing for Serializability 2 Understanding TRANSACTIONS

MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability.

Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. is an official topic in Module 3 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Testing for Serializability 2 Understanding TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, implementation of atomicity and durability, concurrent executions, Serializability, recoverability, implementation of isolation, transaction definition in SQL, testing for Serializability. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

application ഏകദേശം ഏകദേശമായി ഏകദേശം. Technical terms English-ൽ ഏകദേശം
ഏകദേശമായി.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Identify security and integrity threads in database

This module is presented from the official syllabus scope for Non-Relational Database. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify security and integrity threads in database
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Security and integrity threats, Defence 2 Understanding mechanisms

Security and integrity threats, Defence 2 Understanding mechanisms is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security and integrity threats, Defence 2 Understanding mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security and integrity threats, defence 2 understanding mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Security and integrity threats, Defence 2 Understanding mechanisms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- സുരക്ഷയും ആവശ്യതയും, പ്രതിരോധ 2 മനസ്സിലാക്കുന്ന മെക്കാനിസം. നിർവ്വചന/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം. സാങ്കേതിക പദങ്ങൾ ഇംഗ്ലീഷിൽ ഉപയോഗിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

▼ Statistical database auditing & control

Statistical database auditing & control is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Statistical database auditing & control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, statistical database auditing & control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Statistical database auditing & control means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ Security issue based on granting/revoking of 2 Understanding privileges

Security issue based on granting/revoking of 2 Understanding privileges is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security issue based on granting/revoking of 2 Understanding privileges using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security issue based on granting/revoking of 2 understanding privileges should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Security issue based on granting/revoking of 2 Understanding privileges means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- സുരക്ഷാ പ്രശ്നം അധാരത്തിൽ അനുവദിക്കുന്ന/അനുകൂലിക്കുന്ന 2 Understanding പ്രദാനം ചെയ്ത definition/meaning ഉൾക്കൊള്ളിക്കുക. പ്രശ്നം പ്രദാനം ചെയ്ത, steps, application ഉൾക്കൊള്ളിക്കുക. Technical terms English-ൽ ഉൾക്കൊള്ളിക്കുക.

▼ Introduction to statistical database security

Introduction to statistical database security is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction to statistical database security using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction to statistical database security should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction to statistical database security means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇം Introduction to statistical database security ഇം definition/meaning ഇം. ഇം ഇം, steps, application ഇം ഇം. Technical terms English-ഇം ഇം ഇം.

▼ PL/SQL Security - Locks

PL/SQL Security - Locks is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PL/SQL Security - Locks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pl/sql security - locks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PL/SQL Security - Locks means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-PL/SQL Security – Locks ഫലപ്രസാദൻ
 definition/meaning ഫലപ്രസാദൻ. ഫലപ്രസാദൻ ഫലപ്രസാദൻ, steps,
 application ഫലപ്രസാദൻ ഫലപ്രസാദൻ. Technical terms English-ഫലപ്രസാദൻ
 ഫലപ്രസാദൻ.

▼ Implicit locking, types and levels of locks, 2 Understanding explicit locking

Implicit locking, types and levels of locks, 2 Understanding explicit locking is an official topic in Module 4 of Non-Relational Database. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Implicit locking, types and levels of locks, 2 Understanding explicit locking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, implicit locking, types and levels of locks, 2 understanding explicit locking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Implicit locking, types and levels of locks, 2 Understanding explicit locking means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇംപ്ലിസിറ്റ് ലോക്കിംഗ്, ത്വയ്കളും ലെവലുകളും ഓഫ് ലോക്കുകൾ, 2 Understanding explicit locking ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning ഏതെങ്കിലും ഏതെങ്കിലും. ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും, steps, application ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-ഇംപ്ലിസിറ്റ് ലോക്കിംഗ്.

▼ **Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security - Locks - Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html**

Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security - Locks - Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html is an official topic in Module 4 of Non-Relational Database. Study the

terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, oracles' named exception handlers 2 understanding database security security and integrity threats, defence mechanisms, statistical database auditing & control. security issue based on granting/revoking of privileges, introduction to statistical database security. pl/sql security – locks – implicit locking, types and levels of locks, explicit locking, oracles' named exception handlers. text / reference t/r book title/author elmasri r. and s. navathe, database systems: models, languages, design r1 and application programming, pearson education, 2013. r2 adam fowler, nosql for dummies, john wiley & sons, 2015 nosql data models: trends and challenges (computer engineering: databases r3 and big data), wiley, 2018 handbook of database security, applications and trends r4 gertz, jajodia online resources sl. no website link 1 <https://www.w3resource.com/redis/> 2 <https://www.w3schools.in/category/mongodb/> 3 <https://www.tutorialspoint.com/arangodb/index.html> 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Oracles' named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracles' named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources Sl. No Website Link 1 https://www.w3resource.com/redis/ 2 https://www.w3schools.in/category/mongodb/ 3 https://www.tutorialspoint.com/arangodb/index.html 4 https://www.tutorialspoint.com/cassandra/cassandra_introduction.html means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Oracle's named Exception Handlers 2 Understanding Database security Security and integrity threats, Defence mechanisms, Statistical database auditing & control. Security issue based on granting/revoking of privileges, Introduction to statistical database security. PL/SQL Security – Locks – Implicit locking, types and levels of locks, explicit locking, Oracle's named Exception Handlers. Text / Reference T/R Book Title/Author Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design R1 and Application Programming, Pearson Education, 2013. R2 Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015 NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases R3 and Big Data), Wiley, 2018 Handbook of Database Security, Applications and Trends R4 Gertz, Jajodia Online Resources SI. No Website Link 1

<https://www.w3resource.com/redis/> 2

<https://www.w3schools.in/category/mongodb/> 3

<https://www.tutorialspoint.com/arangodb/index.html> 4

https://www.tutorialspoint.com/cassandra/cassandra_introduction.html

മലയാളത്തിൽ നിന്ന് definition/meaning നൽകുക. മലയാളത്തിൽ നിന്ന് definition, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key. (3 marks)**

► **Define or explain Key-Value Data Model Columnar Data Model. (3 marks)**

► **Define or explain Graph Based Data Model. (3 marks)**

► **Define or explain NoSQL system ways to handle big data 2 Understanding problems. (3 marks)**

Module 2

► **Define or explain Data Storage and Indexes- File organizations 2 Understanding.. (3 marks)**

► **Define or explain Primary indexing. (3 marks)**

► **Define or explain Secondary index structures. (3 marks)**

► **Define or explain Various index structures - hash-based. (3 marks)**

Module 3

► **Define or explain Transaction concept, transaction state. (3 marks)**

► **Define or explain Implementation of atomicity and durability. (3 marks)**

► **Define or explain Concurrent executions. (3 marks)**

► **Define or explain Serializability. (3 marks)**

Module 4

► **Define or explain Security and integrity threats, Defence 2 Understanding mechanisms. (3 marks)**

► **Define or explain Statistical database auditing & control. (3 marks)**

► **Define or explain Security issue based on granting/revoking of 2 Understanding privileges. (3 marks)**

► **Define or explain Introduction to statistical database security. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **NoSQL Data model: Aggregate Models- 2 Understanding Document Data Model- Key.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Data Storage and Indexes- File organizations 2 Understanding..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Transaction concept, transaction state.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Security and integrity threats, Defence 2 Understanding mechanisms.**

3-mark explanation: Explain one principle, process or comparison from

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link <https://www.w3resource.com/redis/>
<https://www.w3schools.in/category/mongodb/>
- <https://www.tutorialspoint.com/arangodb/index.html>
- https://www.tutorialspoint.com/cassandra/cassandra_introduction.html

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5303B. Verify institutional updates against the current official curriculum.